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Large Low Background kTon-Scale Liquid Argon Time Projection Chambers

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Abstract: We find that it is possible to increase sensitivity to low energy physics in a third or fourth DUNE-like module with careful controls over radiopurity and targeted modifications to a detector similar to the DUNE Far Detector design. In particular, sensitivity to supernova and solar neutrinos can be enhanced with improved MeV-scale reach. A neutrinoless double beta decay search with ^{136}Xe loading appears feasible. Furthermore, sensitivity to Weakly-Interacting Massive Particle (WIMP) Dark Matter (DM) becomes competitive with the planned world program in such a detector, offering a unique seasonal variation detection that is characteristic for the nature of WIMPs.

1. Introduction

In this study we introduce and discuss a dedicated low background module that would enhance the physics program of next-generation experiments such as the planned Deep Underground Neutrino Experiment (DUNE). Such a low background DUNE-like module could be installed as either module 3 or module 4, the so-called "Module of Opportunity" in DUNE. Such a module would increase the physics reach of supernova and solar neutrino physics, and could potentially host a next-generation neutrinoless double beta decay ($0\nu\beta\beta$) or Weakly Interacting Massive Particle (WIMP) dark matter search. We refer to this design as the Sanford Underground Low background Module (SLoMo).

The physics reach would be enhanced by lowering the nominal energy threshold of a DUNE-like experiment from the anticipated 5-10 MeV to levels necessary to address three potential physics targets, listed in order of increasing difficulty:

- $\sim$ 3.5 MeV energy threshold. This threshold is set by the Q-value of the ^{42}K (daughter of ^{42}Ar) in the detector target. By reducing neutron captures, alpha-emitting radon daughters and pileup events above this threshold, supernova burst neutrino sensitivity could be increased in distance, energy and time. Sensitivity to solar neutrinos would also be enhanced, allowing explorations of interesting solar-reactor oscillation tensions and Non-Standard Interactions.
- $\sim$ 0.5 MeV energy threshold. This threshold is set by the decay Q-value of the ^{39}Ar in the detector target. With reduction of electron and photon backgrounds in the target, particularly if the ^{42}Ar content is reduced through use of underground argon (UAr), sensitivity to low energy solar neutrinos from the CNO process would allow a precision measurement to be made. Such a detector will be sensitive to $0\nu\beta\beta$ search with loading of ^{136}Xe,.
- $<$ 100 keV energy threshold. This threshold could be achieved by enhancing the light collection within the detector and by lowering the ^{39}Ar background by deploying UAr. With rejection of electron recoil backgrounds (using timing based pulse shape discrimination), a sensitive WIMP dark matter search could take place, and interesting phenomena such as a supernova coherent elastic neutrino-nucleus scattering signal (a CEνNS glow) could be studied.

These low background targets are achievable due to the unprecedented size and also the increased radiopurity of the module, allowing significant fiducialization and hence less stringent radioactive background requirements than current world-leading dark matter searches. The light enhancements are achievable with current production techniques. As described in Section 2.2.3, production of UAr at the scale required for a DUNE module is potentially achievable, though it requires a dedicated effort to identify the potential argon source and work with commercial gas suppliers

In this paper in Section 2 we outline the design of the module and discuss potential paths to achieve the detector requirements. In Section 3 we present our initial studies of physics reach of this detector.

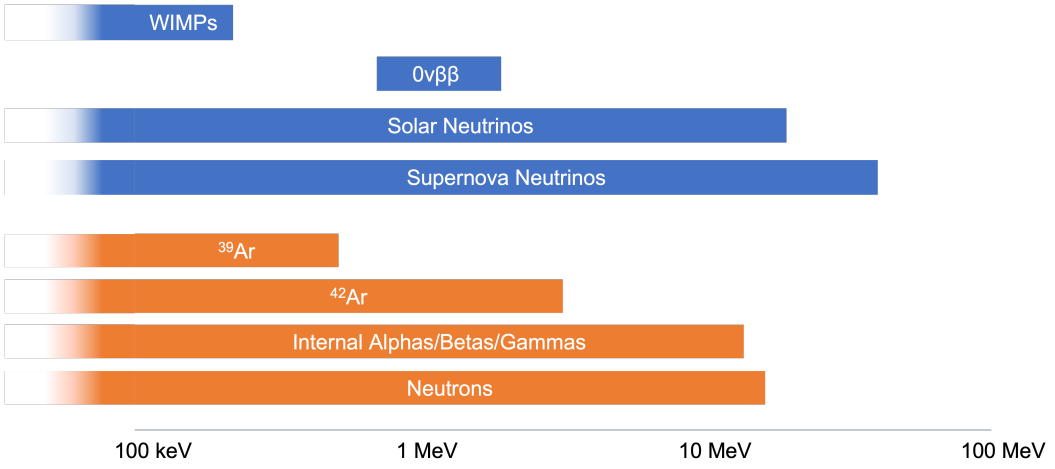


Figure 1. Summary of physics targets of this low background module and the primary radiological backgrounds.

2. Detector Design

This section outlines the proposed design for the low background module and describes in detail the radioactive background control and photon detection system enhancements required to enable physics measurements outlined in the introduction. We note this module design is not necessarily endorsed by the DUNE collaboration, as the so-called "Phase II" process that includes building the final two far detector modules is in its early stages.

2.1. Module Layout

A low background module and its attendant physics goals are enabled most simply by minimizing the detector components in the bulk of the argon. The resulting module must still allow for very good light detection efficiency and for charge detection efficiency similar to existing designs of large LArTPCs [1, 2, 3]. For the benefit of conforming to the longstanding plans for the Long Baseline Neutrino Facility (LBNF) far detector complex and cavern layouts, we also want to use the same commercial cryostat concept and existing module designs to the greatest extent possible and perturb them only where necessary. Starting with the existing DUNE modules, simple modifications assure minimal disruption to the main long baseline neutrino oscillation program to measure remaining parameters in the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix.

2.1.1. Single phase We therefore start with DUNE's Far Detector Vertical Drift Module (VD) [4], sometimes referred to as Module 2, and consider design modifications to suit our low background purposes. We show a working design in Figure 2.

Water in "bricks" which are imagined to be nestled among the I-beam support structure

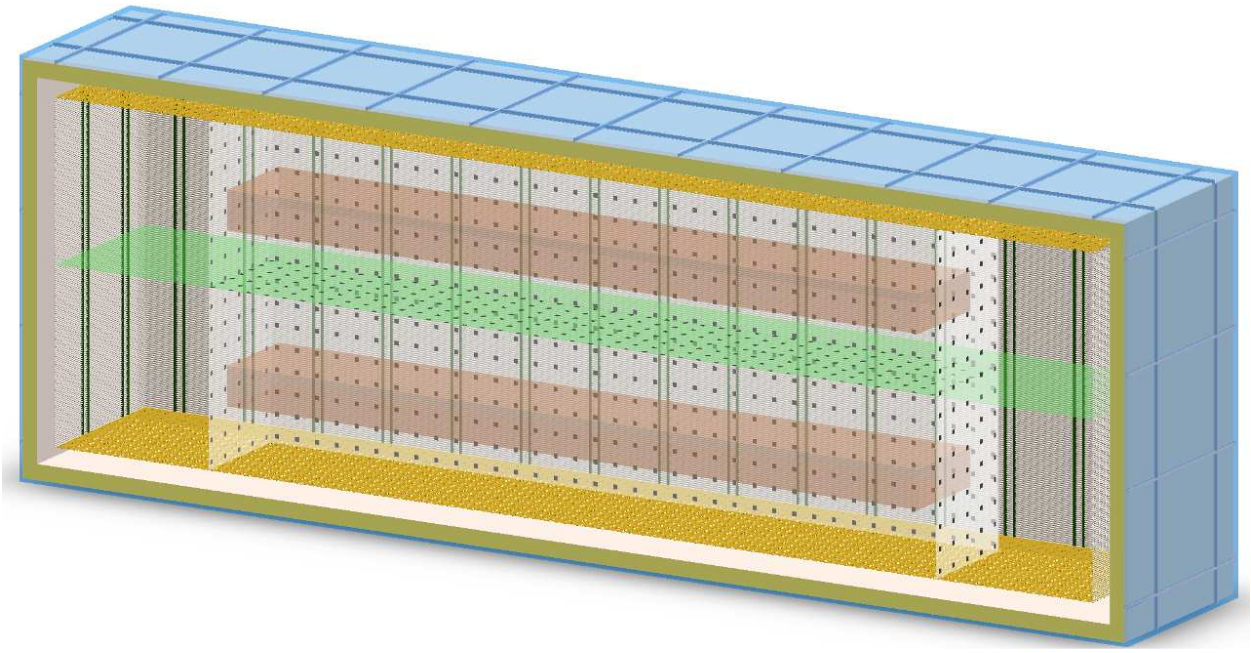
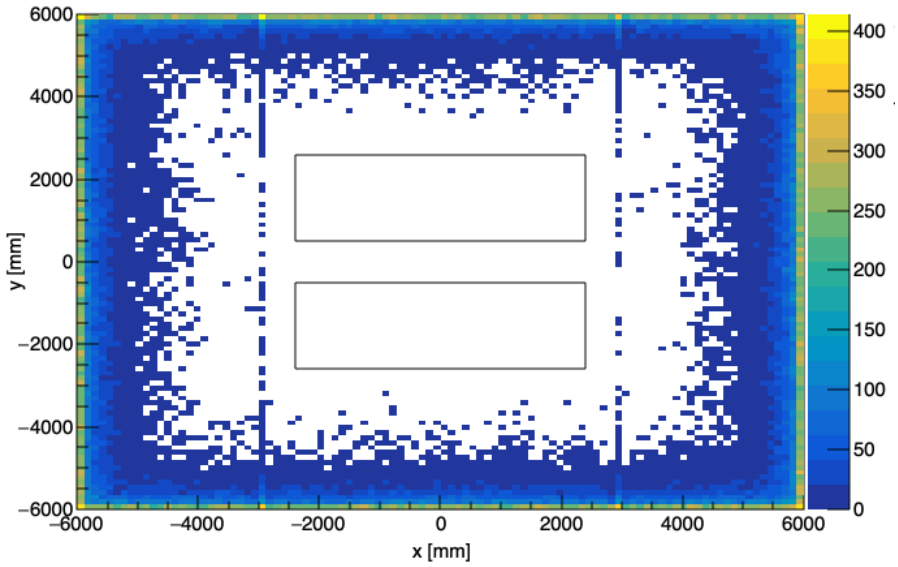


Figure 2. Shown is the base design for the proposed low background detector. Blue shows external water "bricks". The top and bottom yellow planes are the Charge Readout Panels unchanged from the Vertical Detector design. The central cathode is in green. The white box of acrylic (full interior volume) is of thickness 1 inch and has x,y,z extent of 6,12,40 m. The black points are SiPM modules shown here at a coverage of 10%, while some studies in this paper use up to 80% coverage. A proposed fiducial volume totaling 2 kTon is shown in the two beige boxes. This paper also considers a 3 kTon volume.

achieve large external neutron reduction, as discussed in Section 2.2.1. We keep the Charge Readout Planes of the VD unaltered. Similarly the central cathode of the VD is preserved with the exception that the sparse and mostly distant photon detection modules, known as X-Arapucas, are swapped out for the SiPM modules mounted within the cathode plane – at least on that part of the cathode in the inner region of this detector. An acrylic box of one inch thickness with x,y,z extent of 6,12,40 m serves to mount SiPM modules and reflective WLS foils. Here x is the horizontal dimension, y is the vertical dimension, z is in the beam direction. SiPM modules are mounted on the inside of the acrylic box at anywhere from 10-80% area coverage. We envision two 1 or 1.5 kTon long skinny fiducial volumes, depending on the study, that have a stand-off of 1.5m from the central cathode. We also discuss, alternatively, a 3 kTon fiducial volume in this paper in studies where backgrounds from the central cathode are thought to be small or events from it can be reconstructed and cut away.

The bulk volume of this module consists mostly of argon, with only small material contributions such as the slender support structures for the cathode plane panels. As in the VD, there are two 6 m vertical drifts.



(a)

Figure 3. Interactions are shown above 100 keV for neutrons emanating from the cold cryostat stainless steel at $2 \cdot 10^{-10}$ neutrons/cm^3/sec for a 1.4 yr exposure. We show a possible ∼3kTon fiducial volume pair, avoiding the cathode, looking down the beam line. (The vertical bands are interactions in the acrylic.)

2.2. Radioactive Backgrounds

To enable the physics targets for this module, improvements in control of internal and external radioactive background levels are required. Making a module of this size low background will require significant quality and materials controls beyond what has been been attempted by previous experiments, and certainly beyond what is required for the Phase I DUNE program. However, we note that due to the increased size of this module self-shielding in the argon allows the background requirements to be less stringent than those expected to be reached by the current Generation 2 (G2) dark matter experiments. Thus future research and development will need to focus on how to scale the techniques successfully deployed to low background dark matter or neutrinoless double beta decay searches to the kTon scale.

A particular concern for this module will be neutron-induced background events, which will be the main background to the neutrino searches above 3.5 MeV thresholds. A neutron capture in ^{40}Ar produces a 6.1 MeV gamma cascade which can Compton scatter or pair produce electrons which can mimic the charged current neutrino interactions in the argon. Captures on ^{36}Ar can produce 8.8 MeV gamma cascades. Neutrons are also the primary backgrounds for the lowest energy searches for WIMP dark matter, where nuclear recoils can mimic the signal. Below 3.5 MeV the primary backgrounds will come from alpha, beta and gamma emitting isotopes within the argon or detector materials.

2.2.1. Cavern Neutron Backgrounds The most significant source of neutrons will likely be those induced by spontaneous fission or (α, n) interactions from the uranium-238 or thorium-232 decay chains within the surrounding rock and shotcrete of the detector cavern. As reported in [5], the external neutron rate at SURF (a likely hosting laboratory for this low background module) is assumed to be 1.0×10^{-5} n/cm^2/s. As proposed in References [6, 7], it is possible to add water shielding to a DUNE-like cryostat, taking advantage of the space between the structural supports even when the detector is located within a cavern at SURF with limited space around the detector. Those references shows that a 40 cm water shield, located within the support structure, is enough to lower the external neutron rate by three orders of magnitude. We assume this is achievable for this module.

The strategy to control the radioactive backgrounds from the detector components such as the cryostat will have three parts:

- improvements to material selection,
- additional internal neutron shielding,
- and advanced event selections and analysis tools,

with the aim of lowering the internal neutron rate within the detector by at least three orders of magnitude to match the levels of the external rates.

Aside from external neutrons from the cavern, the main source of neutrons in a DUNE-like detector cryostat is likely to be spontaneous fission or (α, n) interactions from the uranium-238 or thorium-232 decay chains in the order 1 kTon of stainless steel that makes up the I-beam support structure. Research and development will be required to lower the internal background rates by the three orders of magnitude required, for example by careful selection of the raw ingredients and/or control of the manufacturing process. It should be noted that the "Generation 2" dark matter experiments expect to reach neutron rates from their steel a further two orders of magnitude beyond this, so the goal is achievable (even if the scale is much larger than previously attempted). Another area of R&D required to support this goal is improvements in knowledge of (α, n) cross sections, as highlighted in a recent IAEA workshop [8].

Another approach to reduce neutron background from the internal shielding within the detector would be by adding higher density rigid polyurethane foam (R-PUF) insulation and/or boron, lithium or gadolinium loaded material layers within the membrane cryostat structure. Our studies show that this could easily reduce the neutron capture rate in LAr by one order of magnitude. One approach, as planned by the DarkSide collaboration, would be to use the additional planes of Gd-doped acrylic to act as a neutron absorber. DarkSide-20k [9] intends to use multiple layers within a ProtoDUNE-style cryostat for their dark matter search. Another design choice might be to take advantage of the existing cryostat but replace some materials such as the insulating foam with a borated version, e.g., to reduce backgrounds from the support structure.

Analysis based cuts can also be used to remove events. For example, with the low threshold of this planned detector, neutron induced multiple scatters could be tagged and rejected. This

takes advantage of the excellent ($\sim$ 10 mm) position resolution of a TPC. Studies [10] to identify dark matter nuclear recoil backgrounds show $\sim$ 30% reduction at 100 keV threshold and $\sim$ 90% reduction at 50 keV, due to the increased probability of detecting an additional scatter at lower thresholds.

2.2.2. Radon and other internal argon backgrounds Radon is an important background that must be controlled as it can diffuse throughout the detector, entering the fiducial volume. The radon decays to a number of daughter isotopes that can be direct backgrounds (for example ^{214}Bi for a neutrinoless double beta decay search) or that can produce neutrons through (α, n) interactions. For this low background module we set a radon target level in the liquid argon of 2 μBq/kg. This is about three orders of magnitude below the expected DUNE radon level of 1 mBq/kg [11, 12, 13]. 2 μBq/kg has been achieved in liquid argon by the DarkSide-50 experiment [14], and exceeded by DEAP-3600 which achieved a level of 0.2 μBq/kg [15]. It should be noted that the higher volume-to-(radon-emanating)-surface ratio in a large detector such as DUNE compared to dark matter detectors will help achieve this target.

To reach this level in a kTon-scale detector will require research and development to implement a combination of the following techniques:

- **Radon removal during purification via an inline radon trap**. No radon removal is in the current design of the purification system for the baseline DUNE. Dark matter and neutrinoless double beta decay search experiments typically use cooled, activated charcoal radon traps to remove radon directly from the recirculating target. The most sensitive dark matter experiments typically purify the argon in the gaseous phase [16, 17], however such an approach would be impractical for a kTon-scale experiment. Borexino used a charcoal radon trap to purify liquid nitrogen [18], and such an approach could be adopted and scaled appropriately for a low background module. New materials such as Metal-organic frameworks could improve the capture-potential beyond charcoal, allowing a potential shrinkage of footprint of a radon-capture facility to fit the existing cavern designs. Recent evidence from MicroBoone [19] indicates that a copper filter purification system similar to that planned for DUNE may remove greater than 97% or 99.999% of the radon (depending on whether slowed or trapped) in the system without the need for additional removal techniques.
- **Emanation measurement materials campaign**. All materials used in detector construction are known to emanate radon at some level. A large-scale emanation assay campaign to identify materials suitable for construction, similar to the QA/QC campaign described above will be required to ensure the detector can meet the target. A topic for R&D will be how to increase throughput of samples, as emanation measurements typically take two weeks per sample.
- **Surface treatments**. For large components such as the cryostat where it may be impractical and costly to make significant improvements to the radiopurity, surface treatments can be used to lower emanation rates. It is known that acid leaching and electropolishing lowers

emanation rates. Coating the inner surface of the cryostat with a radon barrier could lower emanation rates from this significant source.

- **Dust control**. Dust is a significant radon source and cleanliness standards will be higher in this low background module than the baseline DUNE design. Cleanliness protocols and requirements R&D will be necessary to develop automated techniques applicable to the thousands of m^2's of surface area of a DUNE-like cryostat, for example.
- **Radon reduction system during installation and operation**: The mine air underground is radon laden up to 1,000 Bq/m^3 and radon daughter plate-out during installation, filling and operation must be controlled. An upscaled vacuum swing system with large charcoal columns in parallel to remove radon from the ambient air and to provide radon-free air to the cleanroom and cryostat would be suitable. Vacuum swing systems providing radon-free air have been successfully employed by e.g. Borexino [20], LZ [21] and SuperCDMS [22].
- **Drifting of charged daughters to cathode**. Several daughters in the radon chain are charged and will drift towards the cathode and out of the fiducial volume. This effect may be countered by mixing effects of the purification system however.
- **Alpha tagging through pulse shape discrimination**. Alpha events in the radon chain that produce neutrons directly in the argon may be taggable, by identifying the alpha track before the (α,n) event. Though the amount of light may be relatively small, the timing profile is distinct and may allow pulse shape discrimination on this module with enhanced optical systems.

2.2.3. Underground Argon Atmospheric argon (AAr) consists mostly of ^{40}Ar which is stable. There are some long-lived radioactive isotopes-^{39}Ar ($T_{1/2}$=269y, Q_β=565 keV), ^{37}Ar ($T_{1/2}$=35d, Q=813 keV), ^{42}Ar ($T_{1/2}$=32.9y, Q_β=599 keV) [23] which are, in atmosphere, produced primarily by cosmic ray-induced reactions in ^{40}Ar. The use of atmospheric argon in a low-threshold multi-ton scale argon detector has limitations due to high ^{39}Ar activity (1 Bq per kg of argon [24]) in atmospheric argon (AAr). Radiogenic and cosmic-ray muon-induced interactions, especially on K and Ca isotopes, can produce ^{39}Ar underground. The dominant production channels are negative muon capture on ^{39}K and (α, n)-induced (n,p) reactions on ^{39}K. ^{39}Ar production underground decreases significantly with depth[25] as muon flux decreases. DarkSide-50, the only experiment to use underground argon (UAr), measured the ^{39}Ar activity of 0.73 mBq/kg [26], a factor of 1400 smaller than in AAr. With the ARIA project[27], which is planning for a throughput for ^{39}Ar processing of $\sim$ 10 kg/day, the DarkSide collaboration is planning to further reduce the ^{39}Ar present in UAr through large-scale isotopic separation by cryogenic distillation.

^{42}Ar decays in the bulk argon volume will produce ^{42}K isotopes. While the ^{42}Ar beta-spectrum has an endpoint of 599 keV, Betas from ^{42}K-decays span a much larger energy range up to 3.5 MeV, and can be problematic backgrounds. Since UAr should be heavily depleted of ^{42}Ar, significant suppression of ^{42}K decay backgrounds is achievable with UAr. In the atmosphere, the ^{42}Ar concentration is $\sim 10^{-20}$ ^{42}Ar per ^{40}Ar atom [28],[29], which is four orders of magnitude

smaller than the concentration of ^{39}Ar. The daughter isotope of ^{42}Ar, ^{42}K ($T_{1/2}$= 12 h) has two major decay modes : 1) direct beta-decay to the ground state of ^{42}Ca (Q_β= 3525 keV, BR=81 %), and 2) Beta-decay (Q_β= 2001 keV) to an excited state of ^{42}Ca followed by a prompt 1524 keV gamma emission. In the atmosphere ^{42}Ar is primarily produced by $^{40}\mathrm{Ar}(\alpha, 2p)^{42}\mathrm{Ar}$ occurring in the upper atmosphere [29], where energetic alphas are readily available from cosmic-ray muon interactions. Production through two-step neutron capture is also possible but greatly sub-dominant due to the short-lived intermediate isotope ^{41}Ar [30]. The ^{42}Ar production rate underground is not known, but it is expected to be several orders of magnitude smaller than in AAr. Particle interactions on isotopes of K, Ca, and Ti can produce some ^{42}Ar in the earth's crust, given the relatively high abundance of the elements (by mass-fraction[31]:K-2.09%,Ca-4.15%,Ti-0.565%). However, the reaction thresholds are high, making ^{42}Ar production energetically not possible by fission, (α,n)-neutrons or alphas from the natural radioactivity chains of ^{238}U, ^{235}U, and ^{232}Th. Energetic particles from cosmic ray muon-induced interactions can produce ^{42}Ar. But at the depths at which underground argon is usually extracted, the cosmic ray muon-flux should be hugely suppressed, so ^{42}Ar production should be negligible. Based on GERDA's findings[32], following ^{42}Ar decays, ^{42}K nuclei could retain the positive charge long enough to drift in the influence of electric field. So, we expect ^{42}K ions to drift and move towards the cathode plane, which suggests an additional suppression of ^{42}K backgrounds is achievable through fiducialisation.

The ^{85}Kr isotope, predominantly a β-emitter, has a half-life of 10.7 years and Q-value of 687 keV[23]. Primary modes of ^{85}Kr production are spontaneous fission of uranium and plutonium isotopes, neutron capture on ^{84}Kr, and human-induced nuclear fissions in nuclear reactors[33]. We would expect ^{85}Kr to be present at some level in AAr. However, its concentration can vary across argon extraction sites. Using the UAr data, DarkSide-50 measured ^{85}Kr activity of 2 mBq/kg[26], a few orders of magnitude smaller than in AAr. ^{85}Kr concentration in UAr should also vary depending on the location of the gas reservoir and gas origin (mantle-like or crustal-like). DEAP sees no evidence of ^{85}Kr in its AAr after filtering in a charcoal trap with 3.3 tonnes of LAr [34].

We expect argon gas extracted from an underground source to be highly depleted of ^{39}Ar, ^{42}Ar and ^{85}Kr. There is evidence that air infiltration during the UAr extraction could have contributed to the DarkSide-50's ^{39}Ar - actual ^{39}Ar content in the UAr could be significantly smaller (on the order of few tens of μBq/kg)[35]. ^{85}Kr and ^{42}Ar content could also be much smaller. Unlike stable gas isotopes such as ^{40}Ar, which can collect at gas wells over time, isotopes such as ^{39}Ar($T_{1/2}$=269y), ^{42}Ar ($T_{1/2}$=32.9y) and ^{85}Kr($T_{1/2}$=10.7y) diffusing through rocks and collecting in a significant number at the underground gas wells is less likely. However, air-infiltration and cosmogenic activation in the argon bulk could introduce these isotopes in the extracted UAr [36, 37]. Greater care, perhaps, is necessary to ensure avoiding contamination of the UAr during extraction, processing, transport and storage.

While UAr is desirable, it requires a dedicated effort to identify the potential argon source and procure argon on a large enough scale necessary for this project. The Urania plant [9] in southwestern Colorado, USA is expected to produce underground argon from CO_2 gas wells at

a rate of $\sim$ 300 kg/day (at full rate) for DarkSide. The argon source and the production rate is not large enough for a kiloton scale experiment. The authors are in the process of identifying alternative gas wells with enriched argon streams. Discussions with three potential commercial suppliers are ongoing, however the underground source samples are not yet tested and low levels of radioactive isotopes must be proven. Initial gas analysis indicates the mantle origin of this sample, with suppressed cosmogenic production compared to a crust source. Based on estimates by the gas suppliers, the production cost could be as low as three times the cost of atmospheric argon and 5 kTon of argon production per year could be achievable.

2.2.4. Surface storage and spallation issues During above-ground storage of our UAr, before placement into the underground module, ^{39}Ar is produced by cosmogenic neutrons in the reactions: ^{40}Ar(n,2n)^{39}Ar and ^{38}Ar(n,γ)^{39}Ar. Production cross sections for ^{39}Ar have been measured in [38]. ^{42}Ar is produced primarily in ^{40}Ar$(\alpha, 2p)^{42}$Ar reactions [29]. In a previous work [39], to which we refer the interested reader, the cross sections for these reactions were obtained from nuclear reaction codes and confronted with data where they exist. An evaluation of cosmogenic ^{39}Ar and ^{42}Ar production in UAr stored on the surface can be found in [40]. A full study of expected spallation and pileup backgrounds during the detector operation is in reference [6].

2.3. Light Collection Enhancement

The light collection for this low background module will be enhanced to enable two main goals:

- lower the energy threshold and improve the resolution at these lower neutrino energies;
- improve pulse shape discrimination for radioactive background rejection.

In this section we describe the improvements to the light detection system required to enable this.

2.3.1. Light Collection Efficiency Impacts on Energy Resolution Charged particles that traverse the liquid argon deposit energy and excite and ionize the argon atoms. This process results in the electrons recombining with the ions generating unstable argon dimers, which decay and emit scintillation light. If an electric field is applied, a fraction of the electrons will drift away before recombining. These ionization electrons are collected by the anode plane and create the charge signal by removing electrons that would have recombined yields an anticorrelation between the light and charge signals observed in LArTPCs. The small number of ionized argon atoms at low energies means that fluctuations in this recombination process can smear the amount of charge observed to energy deposits. The Noble Element Simulation Technique (NEST) collaboration has explored the precision of LArTPC for 1 MeV electrons as a function of charge readout signal-to-noise ratio and the efficiency of collecting light [41]. They predict that using only the charge signal in a LArTPC, the most precise one can reconstruct the energy for a 1 MeV electron is 5%

and MicroBooNE has validated this prediction [42]. To improve the energy reconstruction, further one needs to include measurements of scintillation light.

The NEST collaboration explored the expected energy resolution enhancements that a LArTPC can achieve by including light signals along with the charge measurement. Ref. [41] shows that the energy resolution for a 1 MeV electron for LArTPCs with different signal-to-noise ratios (SNR) and varying light collection efficiency. In particular, we see that a LArTPC with SNR near 40 needs 50% of the scintillation light to measure energy deposits with 1% precision.

2.3.2. Photosensors As a baseline design our plan is to use 24 cm^2, Darkside-20k style [9], SiPM tiles. 50% quantum efficiency at visible wavelengths. We assume here another 50% wave length shifter (WLS) efficiency from, e.g., TPB on the tile surface, for a total efficiency of 25%. For maximum light detection we envision covering inside the cathode (between the two planes of cathode wires, as will be done in module 2) and the interior acrylic walls at up to 80% coverage. We assume the Module 2 power-over-fiber concerns [4] to be solved to allow our SiPM coverage of the cathode. The resulting number of SiPM modules for 10% coverage – a value which current studies naively show is sufficient for a dark matter search using pulse shape discrimination – is ~ 50000, to be compared to Darkside20k's planned ~ 10000. However, as discussed in Section 2.3.1, to achieve the energy resolution required for a neutrinoless double beta decay search will require significantly more coverage. It is likely possible to optimize the placement of the tiles around the fiducial volume, to reduce the total number required, and work is ongoing to study this.

2.3.3. Reflectors To maximize the light capture in this module the surface of the acrylic box will be coated with a reflector to create a light-tight inner volume. If a PTFE reflector is used, as in DarkSide-50, reflectivities of 97% should be possible.

2.3.4. Argon Purity The baseline requirement for DUNE is < 25 ppm of nitrogen to ensure that photon propagation in the argon is not attenuated. For our simulation studies we assume that attenuation (absorption) lengths of order 50 m are achievable, which corresponds to nitrogen contamination levels of 1-2 ppm [43]. We note that dedicated dark matter experiments have achieved ppb levels of purity.

3. Physics Studies

This section outlines key physics goals of this module and describes the studies that have been performed.

3.1. Argon-39 Studies

Reduction of the rate of ^{39}Ar is desirable for a number of reasons. This 600 keV endpoint beta emitter decays at a 1 Bq/l rate in ordinary atmospheric argon. It may mask low energy physics by creating optical signals that confuse the reconstruction process. It also represents a serious hurdle in the detector triggering considerations.

Trigger primitives, which constitute a 6 PByte/year data source, not necessarily to be stored into perpetuity, are still an onerous data flow to deal with during steady data-taking. That number drops to a far more manageable tens of TBytes if the ^{39}Ar is reduced by a factor of 1400. In reality of course, much of the data budget will perhaps be consumed by low-threshold activity in this detector.

Perhaps the more pernicious practical problem is that ^{39}Ar beta decays may reconstruct as optical ”hits” which may, in turn, comprise ”flashes” which then confuse the charge-light association for reconstructed physics objects – especially at low energy and far from the light detectors. (Hits are simply individual light signals over threshold and the parameters that characterize them, and flashes are collections of hits in narrow ranges of time, hypothesized to be of the same physical origin.) We have performed a study in a Module 1-like environment, not shown here, that gives the not very surprising conclusion that supernova flashes become unambiguously matched with a x1400 ^{39}Ar reduction.

Third, the highly desirable property of pulse shape discrimination (PSD) in Argon which would allow to subtract out the nuisance ^{39}Ar contribution for our, e.g., dark matter search ambitions, is not practicable if pile-up is too intense. We show in [10] and discuss later in this paper that a x1400 reduction of ^{39}Ar just allows for a search in our fiducial volume down to 100 keV thresholds and perhaps lower.

Here we mention that even a x1400 reduction of ^{39}Ar does not allow this detector to get to arbitrarily low thresholds for searches of physics with electronic signatures – as apposed to neutron-like interactions. We expect that reductions by this amount will still leave ^{39}Ar as an overwhelming background to low-rate, low-energy solar processes, for example.

3.2. Simulation

Almost all studies in this paper are carried out in a standalone Geant4[44] simulation. Source code and build instructions are found at Reference [45]. In that simulation is proper isotope decay and neutron physics, along with optical physics. The volume is basically a 10 kTon box of liquid argon, but with a reasonable model of the cryostat walls on all six sides with charge readout planes (CRPs) made of G10 on the floor and ceiling, as in the VD. Further, there is an acrylic box inside, open on top and bottom and tiled with 24 cm^2 SiPM modules at an 80% coverage. SiPM modules with that same coverage viewing both upper and lower volumes also tile the central cathode plane. See Figure 2 for a representation of our simulated geometry. We use an after-the-fact 25% total

quantum efficiency by merely scaling by that fraction of the detected hits. There is no SiPM electronics response applied. We include a 96% reflectivity of the acrylic surface and a 44% reflectivity for the CRPs (as the holes in each CRP are about 56% of the surface area), and impose an argon attenuation (absorption) length of 50m, and a Rayleigh scattering length of 90cm. All simulations generate their 128 nm photons ab initio from the charge particles which create them and are propagated on the fly, with no lookup libraries, until their end point on a SiPM where they are counted or they disappear through absorption.

3.3. Optical Studies

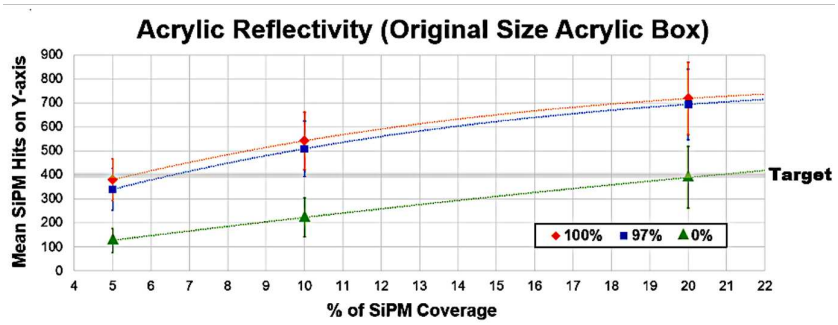
The simulation described in Section 3.2 was used to study the minimal requirements for the optical system to allow pulse shape discrimination in a dark matter search, including the required reflectivity of the acrylic box and anode readout surfaces as a function of SiPM tile coverage. The minimal photon counting requirement for the optical system was set at 400 photons reaching the SiPM surface, which results in a total of 100 photons being detected due to the assumed 25% efficiency described in Section 2.3.2. The 400 (100) photon requirement was chosed as the minimal amount of photons required to perform a pulse shape discrimination analysis such as described in Section 3.7.

The results of the simulation are shown in Figure 4. The studies were performed varying acrylic and anode reflectivity between 0 % and 100 % (a realistic 97 % is highlighted) and then varying the SiPM coverage on the walls and cathode plane to count the accepted photons. Both the standard acrylic box described above and also a maximally sized box where the walls are moved out to the edges of the detector were simulated. This large box would be the worst-case optics scenario, maximising the path length of the photons. The studies show that a relatively modest amount of SiPM coverage of 10-20 % is required even in the worst-case scenarios to reach the target.

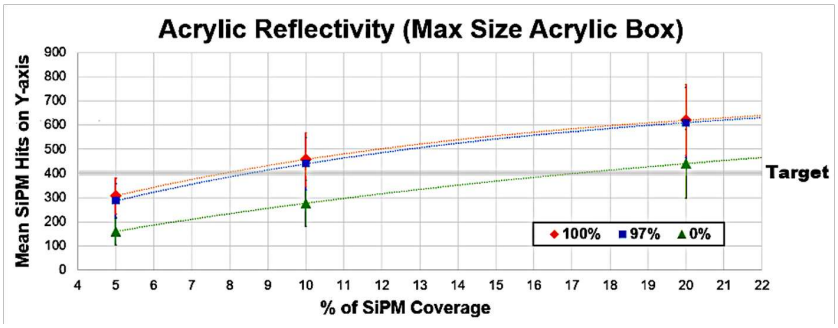
The effect of attenuation with the liquid argon was also studied. Figure 5 shows the number of photons detected at the SiPMs as a function of SiPM coverage for a variety of different attenuation lengths. Assuming the relation between nitrogen contamination of the argon versus attenuation found in [43], this study shows that the 10-20% SiPM coverage is sufficient to tolerate 0.5-5 ppm levels of nitrogen within the argon.

3.4. Supernova Neutrino Physics

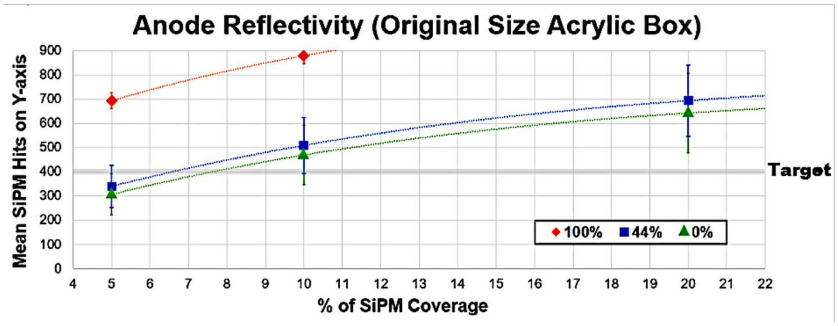
In this section we present several supernova neutrino burst studies that could be enhanced by this module. This includes increased sensitivity to lower energies, later times and sources from greater distances. We also discuss the CEνNS glow sensitivity.



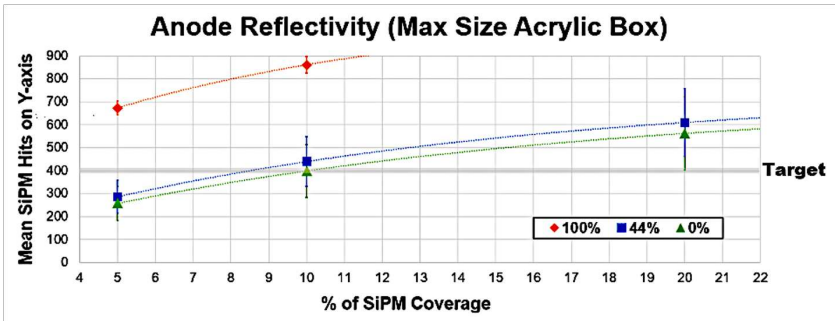
(a)



(b)



(c)



(d)

Figure 4. Number of photons detected at the SiPMs as a function of coverage, varying both reflectivity of the surrounding acrylic and anode plane walls, and the size of the acrylic box containing the inner volume, from Original Size (6x12x20 m^3) to Max Size (at fiducial boundaries at 12x12x60 m^3).

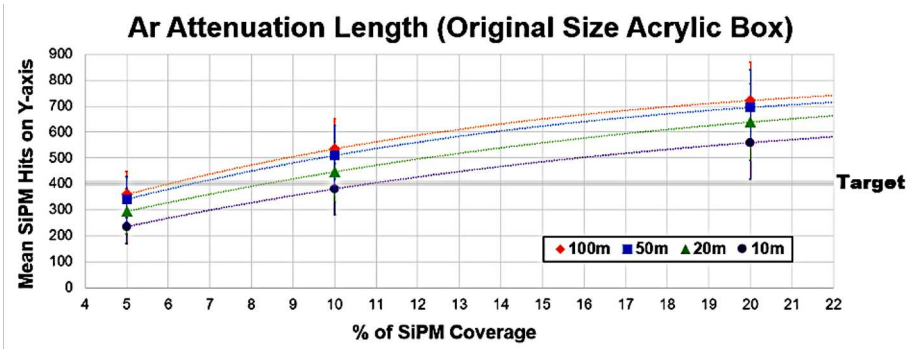


Figure 5. Number of photons detected as a function of SiPM coverage when varying the attenuation length within the argon.

3.4.1. Supernova Energy Spectrum We wish to explore the potential improvement provided by a low background large LArTPC to that achievable in current DUNE Far Detector designs for sensitivity to supernova explosion detection. To this end we simulate a ten-second exposure of our detector to a flux of electron neutrinos from the Livermore [46] supernova model and run them through the MARLEY [47] event generator to produce the final state particles in liquid argon. MARLEY considers all candidate ν_e processes, including coherent, elastic, and charged current processes. The detector response is provided by the simulation described in previous section 3.2.

Using that simulation we count SiPM hits for an 80% coverage and convert to energy using a simple conversion. That conversion comes from running 3 MeV electrons uniformly through the detector and counting SiPM hits in a coarse x,y,z binning of the production point. We emphasize that this study is simplified and that refinements including TPC charge response as well as better SIPM energy resolution will result in improvements. Events must originate in the inner 3 kTon fiducial volume. The result is shown in Figure 6. The most evident feature is that the elastic scattering component of the ν_e flux becomes dominant at low energies inaccessible to the baseline DUNE far detectors – and it does so because neutron rates are required to be low from the cold cryoskin stainless steel and the ^{42}Ar is at very low levels in UAr. The threshold here can go all the way down to 600 keV, which is a factor of roughly 18 lower than the baseline DUNE far detector design.

The low detector threshold allows access to a significant number of elastic scatter events within the liquid argon. This opens up the possibility of reconstructing the position of the supernova with a pointing analysis. Liquid argon TPCs, with the excellent track resolution, are well suited to making this measurement. With the expected reduction in backgrounds in this sample, a clean elastic scatter sample will dominate at thresholds below 5 MeV (see Figure 6), though pointing with these reconstructed tracks is challenging.

3.4.2. Supernova Trigger The trigger system in a DUNE-like LAr detector relies on the so-called Trigger Primitives (i.e. hits, TPs) generated from the electronics connected to the wires or photo-

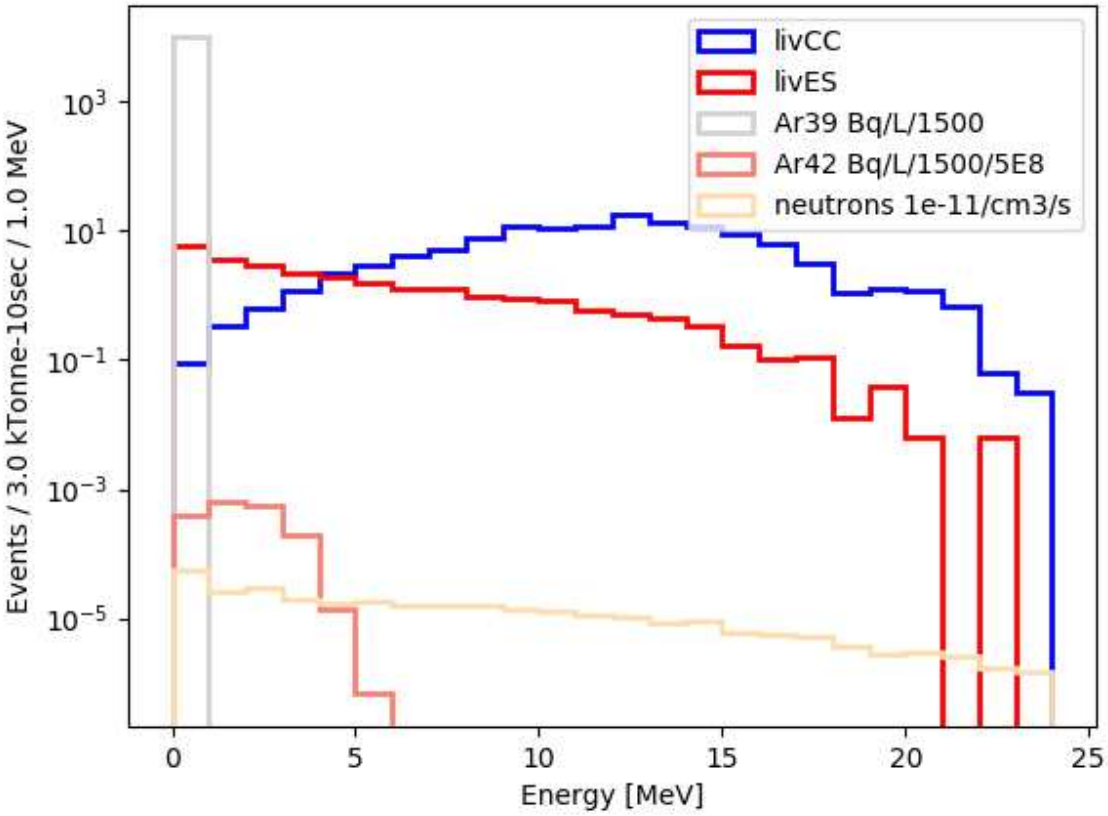


Figure 6. Energy deposited from neutrinos from supernova located at 10 kpc, assuming the Livermore model, during a ten second detector exposure window. Here we presume no Rn222 contribution and a likely too conservative x500 Ar^{42} suppression in UAr with respect to that achievable in atmospheric argon. Reaching 5 MeV SN detection is straightforward in this module.

sensors. They are simple objects constructed from the signal waveform. A stream of TPs arrives in the trigger module of the DAQ, which will use them to form a Trigger Activity (i.e. a cluster of hits, TAs), which is an association in time and space done by a trigger algorithm. A TA is related to each sub-module/component of the detector (e.g. an APA module), and multiple TAs form a Trigger Candidate (TC). Having a positive TC, the readout system stores the requested data. For low energy physics, which does not have an external trigger like beam events, it is essential to understand the detector backgrounds (e.g. radiological and electronics noise), to avoid triggers issued by undesired data. Thus, a Low Background LAr detector can perform better than the current designs.

The bottleneck for designing efficient supernova neutrino burst (SNB) trigger algorithms is the data transfer and storage resources available for the detector. To get the most of an SNB event, around 100 seconds of full detector readout is desirable, which means about 150 TB of raw data for a 10 kTon LAr detector. It will take about one hour to transfer the data from this trigger event from

the detector caverns to the storage on the surface and several additional hours to transmit those data to the primary storage centre. Therefore, while the effective threshold must be set low enough to satisfy the requirements on SNB detection efficiency, it is crucial to not fire too frequently on background fluctuations. A requirement on the fake trigger rate of once per month is determined by these limits on data-handling, for a DUNE-like LAr detector. Additionally, the triggering decision needs to be made within 10 seconds since this is the typical amount of data buffered in the DAQ. SNB triggers are likely to be both TPC- and Photon Detection System (PDS)-based in far detector modules one and two, though at the lower energy thresholds we concern ourselves with in this paper a PDS-only trigger will be required.

In current studies both TPC and PDS information gives a tagging efficiency of about 20-30% for a single neutrino interaction [48]. Reducing the estimated neutron capture rate in the LAr volume by a factor of ten (which is the principal background for SNB triggering, given the higher de-excitation gamma energy of about 6 MeV when compared to other radiological components), this efficiency improves to 70%, which translates to a 100% (20%) SNB triggering efficiency for a Milky Way (Magellanic Clouds) SNB. The trigger strategy described above is a "counting" method. If we utilise the integrated charge of each TP, it is possible to construct a distribution of the TAs raw energy (SNB signal with backgrounds) and compare it to the background-only one. The "shape" triggering method improves the efficiency to tag Magellan Clouds' SNB. The efficiency figure for such a SNB producing ten neutrino interactions (see Ref. [48]) shows the efficiency as 70% in a ten kTons DUNE-like LAr detector using the standard DUNE background model described in Reference [1] and a shape triggering algorithm that keeps the fake trigger rate to one per month.

With the Low Background LAr detector, a less stringent selection can be used, increasing the signal efficiency while keeping the fake trigger rate at the requirement level. Thus, higher efficiencies will be reached with a lower number of SN events, it then being possible to achieve 100% efficiency to trigger a Magellan Cloud SNB. Identifying Andromeda's SNBs is more challenging even with this design since they do not produce enough interactions in the whole detector volume in a 10 seconds window (see the supernova sensitivity plot in Ref. [48]), and lowering the TA requirements would encounter the ^{39}Ar activities.

3.4.3. Late Time Supernova Neutrinos The neutrino flux from a core-collapse supernova is expected to cool over a few tens of seconds, with the late time events getting lower and lower in energy. The tail end of the burst, where a black-hole-formation cutoff may be present, will be challenging to observe. A lower energy threshold extends the time range with which a large liquid argon detector can follow the evolution of the supernova burst [49].

3.4.4. Pre-supernova Neutrino Signal Another benefit of lowering the threshold is potential sensitivity to presupernova neutrinos [50, 51, 52, 53, 54, 55, 56]. The final stages (hours to days) of stellar burning before the core collapse are expected to be associated with an uptick in neutrino

production and energy, and observation of these could provide a true early warning of a core-collapse supernova. The presupernova flux is expected to be small, and energies are typically less than 10 MeV; nevertheless they may be observable in a large liquid argon detector with low threshold [53] for progenitors within a few kpc nearing the ends of their lives.

3.4.5. CEvNS Glow Coherent elastic neutrino-nucleus scattering (CEvNS) [57, 58] is a process that occurs when a neutrino interacts coherently with the total weak nuclear charge, causing the ground state nucleus to recoil elastically. The cross section is large compared to the inelastic charged- and neutral-current interactions, but resulting nuclear recoil energies are in the few tens of keV range. CEvNS has now been observed in argon by the COHERENT collaboration using the stopped-pion neutrino flux from the Spallation Neutron Source [59] with a cross section on order $22 \cdot 10^{-40}\text{cm}^2$ for an incoming average flux of $< E_\nu > \approx 30$ MeV from pion decay at rest.

In a large LArTPC there will be a high rate of CEvNS in a core-collapse supernova burst—approximately 30-100 times more events with respect to ν_eCC (the dominant inelastic channel), depending on expected supernova spectrum. However, each event is individually not likely to produce more than a few detected photons, and sub-50-keV thresholds need to be achieved to find these events, if they are to be found one by one. Even with depleted argon, the ^{39}Ar rate down in this range in our proposed detector is by far overwhelming. However, an alternative is to identify a "CEvNS glow," [60] in which the excess rate of detected photons can be tracked statistically above the ^{39}Ar rate as a function of time in a characteristic explosion. Fig. 7 shows that simulated supernova-induced activity from a burst at 10 kpc over 10 seconds in an inner fiducial 3 kTon gives three distributions: broadly-distributed-in-time, higher-energy charged-current (CC) component, a burst of low-hit-multiplicity CEvNS events at about 0.01 sec, and then the absolutely flat ^{39}Ar activity. In ongoing work, we propose to subtract the reconstructed CC events and fit to the CEvNS bump above the background. We note that in principle, an excess of collected ionization from CEvNS events is observable as a "CEvNS buzz" in coincidence with the CEvNS photon glow.

3.5. Solar Neutrino Physics

In this section we present our enhanced sensitivity to solar neutrino searches including lower energies and enhanced oscillation sensitivity to Δm^2_{21}. This allows exploration of solar-reactor oscillation tensions and Non-Standard Interactions. It also allows a precision CNO solar neutrino measurement and a measurement of the ^{3}He+p solar flux. A first study of DUNE as the next-generation solar neutrino experiment is presented in reference [7]. That study is dominantly one of higher energy ^{8}B CC processes, but in this section we widen the discussion and point out what a lower threshold would offer to solar neutrino studies.

3.5.1. Low Threshold Gains and Elastic scattering Figure 8 shows the number of expected ES interactions over threshold for a 3-kTon·year exposure. One sees that reducing the threshold to 1

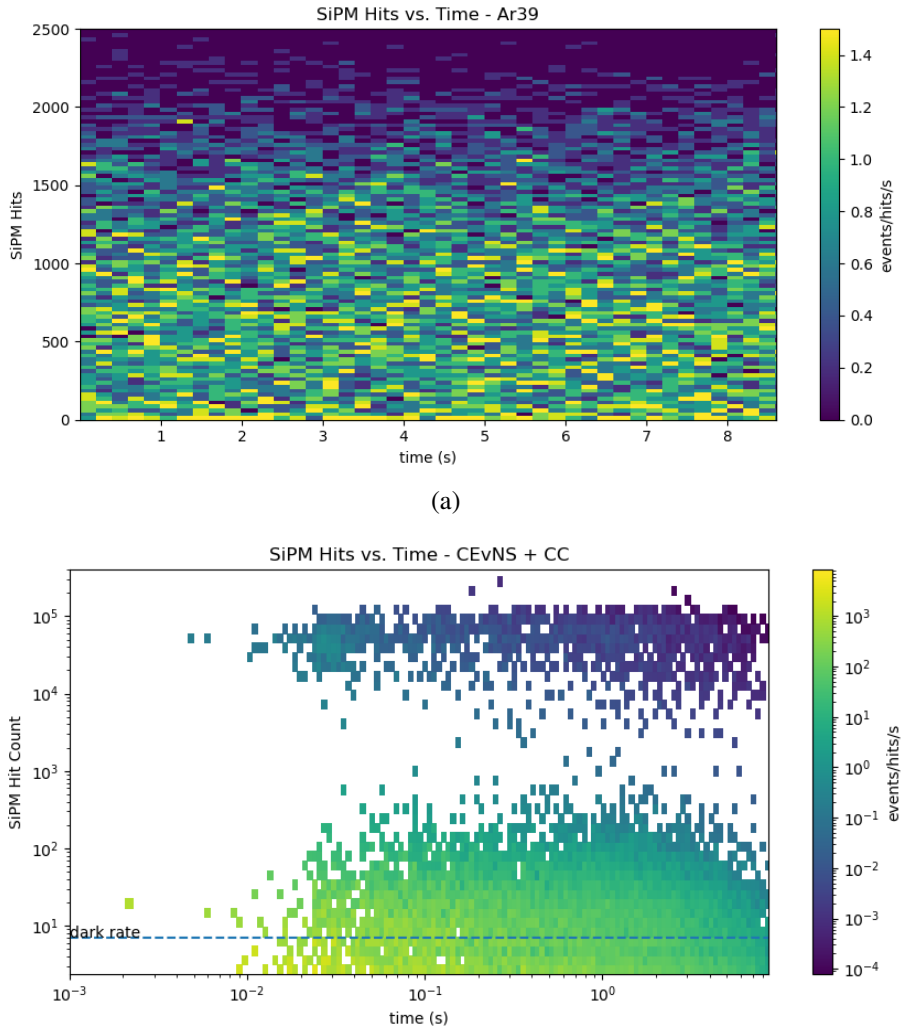


Figure 7. The CEvNS glow is the statistically significant increase observed from the low SiPM hit-count CEvNS Supernova events, shown in bottom population of figure (b) that sits on top of the rate of ^{39}Ar seen in figure (a).

MeV makes *pep* and CNO neutrinos observable. A threshold of 0.5 MeV could add ^{7}Be neutrinos, and 0.1 MeV could allow for detection of pp neutrinos, though this threshold encroaches into the large ^{39}Ar background, even in UAr. The total number of available neutrinos is important for possible studies discussed in Section 3.5.3. This amounts to $9{,}200$ neutrinos for a 1-MeV threshold, $130{,}000$ neutrinos for a 0.5-MeV threshold, and $820{,}000$ for a 0.1-MeV threshold.

In addition to the previously discussed methods for reducing backgrounds, we are investigating directionality with ES for all solar neutrinos and Cherenkov radiation for more energetic ^{8}B neutrinos as ways to enhance neutrino signal over backgrounds.

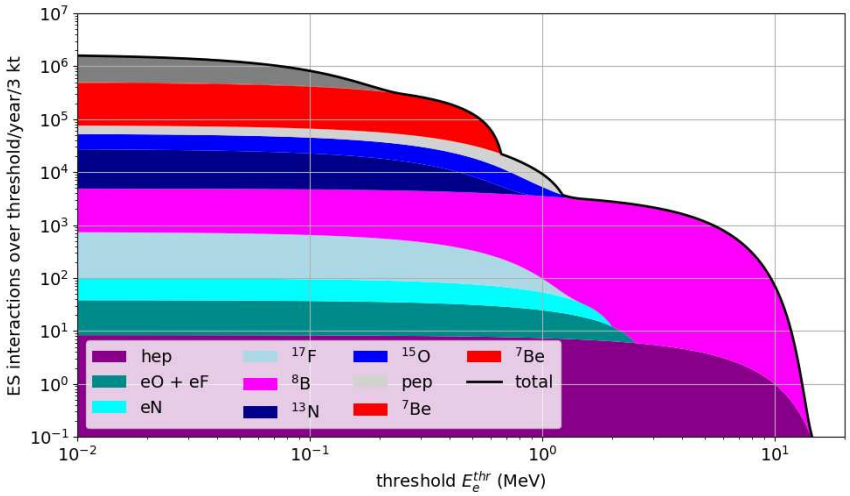


Figure 8. Number of events over threshold for solar-neutrino ES interactions in argon for a 3-kTon·year exposure with contributions from different solar fluxes.

3.5.2. Solar Neutrino Oscillations Current measurements of the neutrino mixing parameter, Δm^2_{21}, using solar neutrinos from SNO and Super-Kamiokande are currently discrepant at 1.4 σ with measurements from KamLAND using neutrinos from nuclear reactors [61]. Differing results from these two methods would suggest new physics possibly involved with exotic matter effects as the neutrino passes through the Sun and Earth. Further data from DUNE will further investigate this discrepancy with high statistical significance. The sensitivity comes from the "day-night" effect, a partial regeneration of the ν_e solar flux due to matter effects in Earth which depends on Δm^2_{21}, neutrino energy, and nadir angle. This is an advantageous strategy allowing for constraints of uncertainties using daytime data.

Solar neutrinos are much lower energy than typically observed in DUNE making reconstruction of these events challenging. Also, at these low energies, radiological backgrounds, principally neutron capture with a contribution from $^{40}\mathrm{Ar}(\alpha,\gamma)$, dominate analysis backgrounds. A low-background DUNE-like module with enhanced light collection will help with both of these challenges.

Improved energy resolution from increased photodetector coverage would significantly improve DUNE-like sensitivity to Δm^2_{21}. This would both improve reconstruction of the dominant neutron background around the 6.1 MeV total visible energy of the neutron capture on argon-40, and better measure the dependence of the ν_e flux regeneration on neutrino energy. A single 10-kTon, low-background module could discern between SNO/SK and KamLAND best fits at over 6 σ. Lower neutron background levels would also improve the signal-to-background ratio for the measurement, which could also lower the energy threshold for detecting and analyzing solar neutrino events. An estimate of sensitivity to Δm^2_{21} with 100 kTon-yrs of exposure with a low-background module is compared to DUNE's sensitivity with 400 kTon-yrs of data from

nominal, horizontal drift modules in Fig. 9. We use a larger 10-kTon fiducial volume for the reduced background/threshold curves in that figure, increased from other studies in this paper.

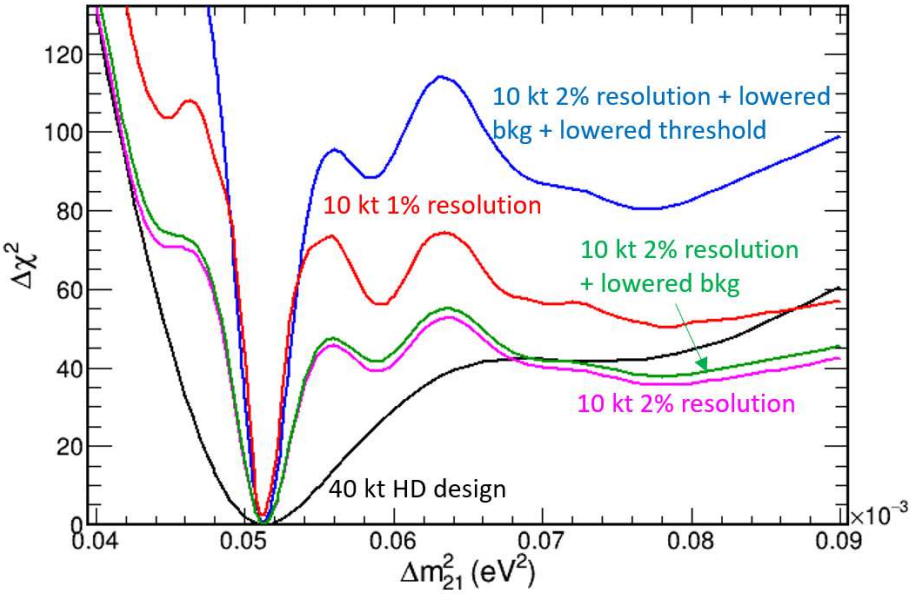


Figure 9. Sensitivity of a low-background module to the neutrino mixing parameter Δm^2_{21} assuming a true value of the solar best fit, $5.13{\cdot}10^{-5}$ eV^2 [61]. Colored contours show sensitivity after 100 kTon-yrs for various detector configurations compared to 400 kTon-yrs of DUNE data with horizontal drift design, shown in black and dominated by CC events.

3.5.3. Non-Standard Neutrino Interactions Non-standard neutrino interactions (NSI) could modify neutrino oscillations in the Sun and result in a different number of neutrinos observed compared to the one predicted by the Standard Model [62] (see also studies with different parameter definitions in [63] and [64]).

The NSI Hamiltonian (for neutral currents only) relevant for solar-neutrino oscillations can be written in the following form:

$$H_\nu^{NSI} = \sqrt{2} G_F (n_u + n_d) \begin{pmatrix} -\epsilon_D & \epsilon_N \\ \epsilon_N^* & \epsilon_D \end{pmatrix}, \quad (1)$$

where G_F is the Fermi constant, n_u and n_d are the up- and down-quark densities, respectively, and ϵ_D and ϵ_N are the diagonal and off-diagonal NSI couplings. These couplings affect ν_e solar survival probability (see Figure 10). The diagonal coupling can mimic different vacuum Δm^2 values, resulting in its incorrect measurement when NSI are not taken into account.

Detection of ES in the proposed module (see Section 3.5.1) allows for a great opportunity to have an almost NSI-independent anchor point in the oscillation probability near 0.1 MeV in addition to investigating NSI in solar-neutrino oscillations at several MeV energies where changes in oscillation probability could be large but not much experimental data exists, yet.

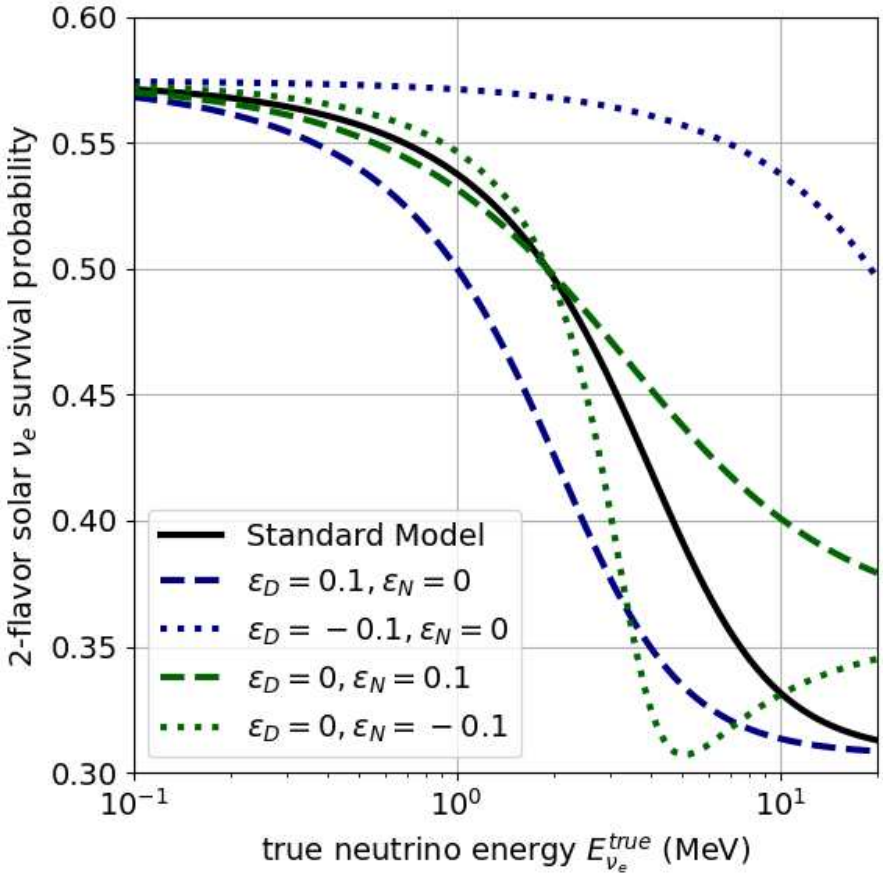


Figure 10. 2-flavor electron-neutrino survival probability for solar neutrinos for the Standard Model and several different NSI couplings.

An example of possible constraints on the NSI couplings is shown in Figure 11. This plot was obtained with the assumption of no backgrounds or systematic errors, an energy threshold of 1 MeV, and an exposure of 3 kTon·years. For comparison, see the larger constraints using current neutrino data available in [61].

3.5.4. Precision Measurement of CNO flux The CNO flux has been measured [65] recently by the Borexino collaboration to 3.5σ above 0, though it yields indistinct information about the high or low metallicity solution. Per [65], the CNO neutrino flux scales with the metal abundance in the solar core, which probes the initial chemical composition of the Sun at its formation. The metal abundance in the core is decoupled from the surface by a radiative zone, and CNO neutrinos are the only probe of the initial condition.

Here we want to investigate if an energy window exists where a precise measurement of the CNO flux can be performed in our low background module. We use our by-now standard simulation tools to count true deposited energy for a variety of backgrounds and solar neutrino

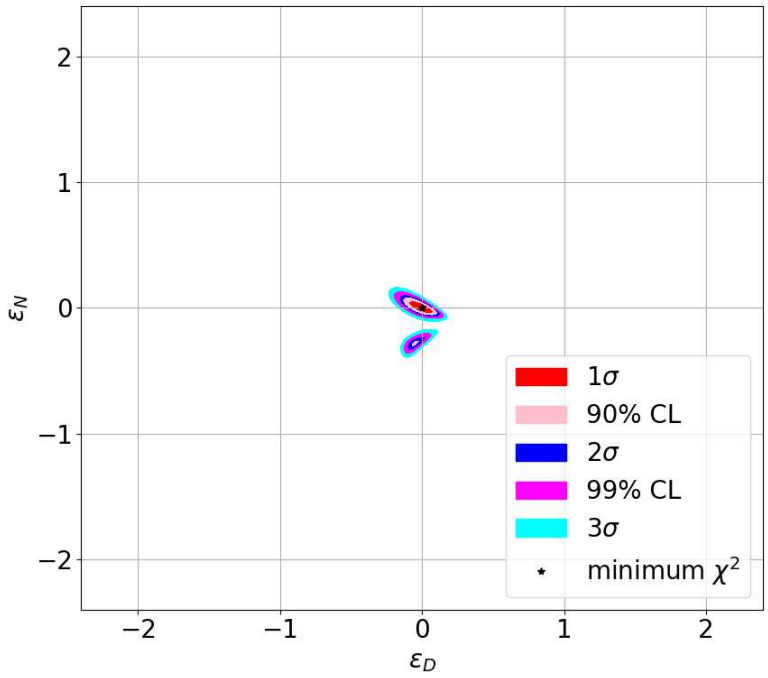


Figure 11. Possible NSI constraints for 3 kTon·years obtained with the proposed module.

sources in a 3 kTon fiducial volume. We impose a 2% energy smearing, as is reasonable by arguments in section 2.3.1. Fluxes come from [66] with a 0.3 survival probability applied, as appropriate to this low energy range. We use the assumption that the ^{42}Ar content will be at a rate that is reduced by a further factor 100,000 compared to atmospheric argon. We emphasize again per section 2.2.3 this is likely very conservative. The result is we see in a window about 0.2 MeV wide just above the *pep* cutoff that the CNO signal sticks up above other solar neutrino sources. This is merely an illustration; a real analysis will likely do a templated fit above the ^{7}Be peak. Neutrons from the cold cryoskin stainless steel are forced to be low, as usual; radon is taken as controlled. The ^{210}Bi background which Borexino took exquisite care to control and measure [65] is yet to be thoroughly investigated here. Nevertheless, there would appear to be a window where the high and low metallicity solutions are statistically separable. By comparison, CNO sensitivity in the two-phase LArTPC program, which studies are further along than the current work, can be seen in [67].

Triggering for CNO neutrinos Initial studies into measuring the CNO flux with TPC triggering are underway, taking into account the full complement of radiological backgrounds predicted in the DUNE detector. In these early studies, TPC triggering requires sufficiently large, coincident clusters on the collection plane and at least one induction plane. We expect future work will in fact lead to a much higher-efficiency, light-based trigger being implemented for most work discussed in this paper. The dominant background in the 1.4 - 2.0 MeV energy window is radon in this

early TPC triggering study, not in fact solar neutrinos – despite the rate reduction by a factor of 10^3 predicted in the low background module. In order to perform the true CNO detection there is clearly much work to be done in offline processing to accurately characterize and implement rejection by alpha detection, Bi-Po coincidence, disproportionate activity near the cathode from drifting cation decay products, and emanation properties of the various detector materials.

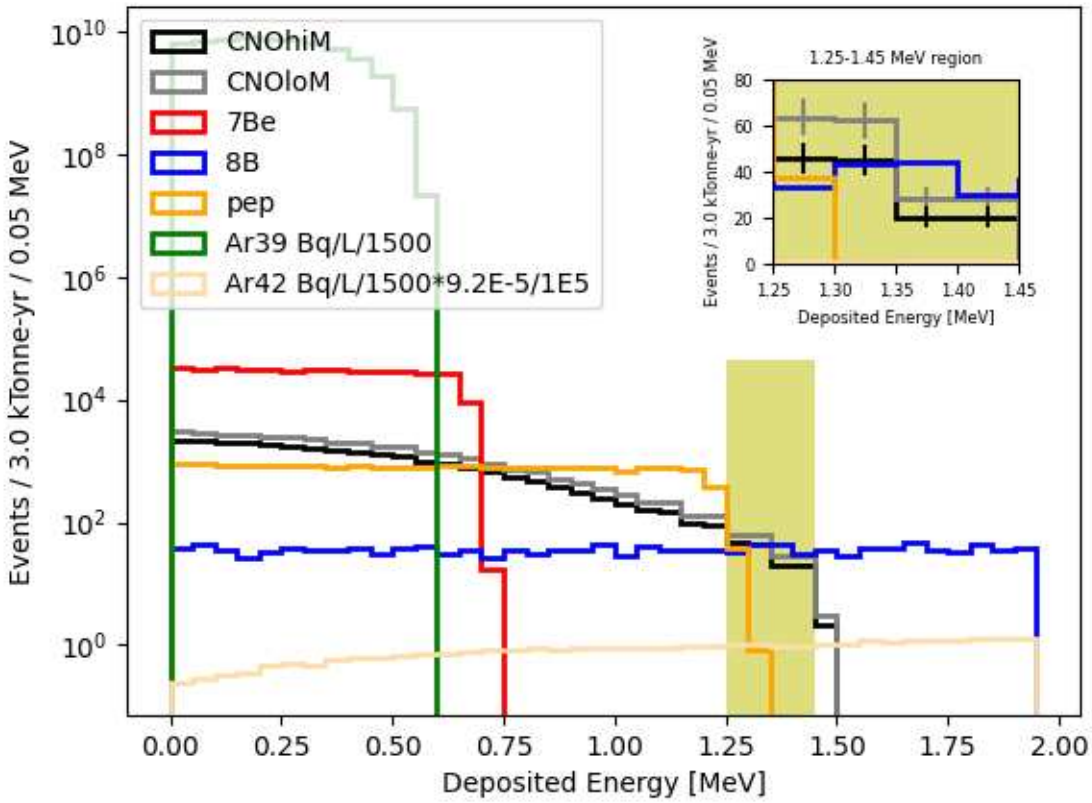


Figure 12. CNO solar neutrinos with backgrounds taking radon to be solved and negligible and neutrons constrained. This is 2% smeared true deposited energy in our inner 3 kTon fiducial volume in a year. Note that a likely, low ^{42}Ar level is shown that reveals that a CNO fit is possible above the ^{7}Be peak and explicitly by a simple counting experiment in roughly the region shown in yellow. This is a lower ^{42}Ar level than shown in Fig 1, but still realistic. The inset zooms in on the 1.25-1.45 above the *pep* region to show that the signal statistics are large enough to favor either high- or low-metallicity after a year.

3.5.5. Precision measurement of hep flux The hep solar neutrino process ($^3\mathrm{He} + p \rightarrow ^4\mathrm{He} + e^+ + \nu_e$) produces the highest energy neutrinos, though they have the lowest flux and have not yet been observed. This low background module will be able to measure tens of these neutrinos per year via CC events, with a significant reduction in background due to radon and neutron interactions within the liquid argon.

3.6. Neutrinoless Double Beta Decay

A discovery of neutrinoless double beta decay ($0\nu2\beta$) is the most straightforward way to prove the Majorana nature of neutrinos. It would be parsimonious to be able to search for $0\nu2\beta$ in the same detector we are suggesting to use for the other measurements mentioned in this paper. ^{136}Xe is one isotope in which much work is being performed worldwide to try to make this discovery [68, 69].

Loading large LArTPCs with few-percent level Xe and measuring neutrinoless double beta decay has been suggested in [70] and [71] among other places. In this subsection we show that naively a discovery is likely possible with a signal ^{136}Xe half-life of $5 \cdot 10^{28}$ years, and is quite apparent at $1{\cdot}10^{28}$ years, provided energy resolution requirements can be met. We imagine, say, a five-year search campaign for $0\nu2\beta$ at the end of the prosecution of the baseline physics program of this module, since any dark matter search requiring PSD would necessarily be compromised by xenon loading.

We start with a $0\nu2\beta$ signal calculated using a Gaussian 1.5 (3) % energy resolution at 2.435 MeV with $1/\sqrt{E}$ dependence for top (bottom) plots in Figure 13. Energy resolution ambitions are consistent [72] with what we may expect in a LArTPC from charge energy collection in combination with the photon readout system, as well as discussion in section 2.3.1. We similarly smear the 2ν double-beta true spectrum by these resolutions. And for the ^{208}Tl background emanating from the G10 of the charge readout planes we use only the expected rate from the simulation, creating the actual spectrum by likewise smearing the 2.6 MeV gamma by 1.5 (3)% in top (bottom) plots. For the solar and ^{42}Ar backgrounds previously discussed, on the other hand, we use the resolution from the simulation that uses only the poorer light-only collection from the SiPM hits. These are flat backgrounds which extend to low energy where signal will likely be obscured in the noise of the charge readout, hence the need to rely on only the SiPMs. For both plots we use a 3% concentration of ^{136}Xe in our inner volume, using a 2.0 kTon fiducial volume, and propose to run for five years. We plot a signal corresponding to a $(5)1 \cdot 10^{28}$ year $0\nu\beta\beta$ half-life.

Among the assumptions made for our figures is that underground argon can give a ^{42}Ar suppression of a factor of 10 beyond that in atmospheric Ar. See section 2.2.3 which indicates this is likely easily achievable, up to processing concerns. We also take in Fig 13 that the irreducible solar ^{8}B elastic scattering may be suppressed by a factor of two (conservatively down from three assumed in Reference [73, 74]) from inspecting Chernkov/Scintillation light ratios in single versus double electron events. We impose no efficiency hit due to this cut, whereas Reference [73] uses an efficiency of 0.75. More careful event reconstruction studies are needed to bear out the reasonableness of this cut. We assign the ^{208}Tl in the G10 charge collection planes a Th concentration of 50 mBq/kg. Charged current solar neutrino events, are in principle reducible to zero, due to the excited state gammas that are emitted. And similarly, neutrons shall be almost entirely removed using pulse-shape discrimination. We again take the radon issue to be solved for the sake of this study. A 2σ band to either side of the $0\nu\beta\beta$ energy of 2.435 MeV is shown.

We take $\mathrm{E}_{res} \sim 1.5$ % in the top plot at the Q value, even though, as we have said, it is not

immediately obvious we can achieve that (nor in fact are we currently confident we can reach the 2.5% of the bottom plot) with our charge readout plus 80% SiPM coverage, open as it is at the top and bottom. That study is beyond the scope of this paper. However, we see that there is sensitivity in this detector to $0\nu\beta\beta$ discovery at lifetimes that stretch the reach of coming experiments, despite the large, irreducible solar ^{8}B neutrinos which elastically scatter off the mostly argon target.

3.7. Dark Matter

It is known that a large amount of dark matter exists within the Universe, that has so far only been observed by gravitation interactions. One popular candidate for the dark matter is the Weakly Interacting Massive Particle, or WIMP, that is the focus of several current and future experiments. The potential of using this low background module to search for WIMP dark matter was studied in Reference [10]. This study assumed a dual phase TPC design, with a single fiducial volume at the center of the detector (rather than the split fiducial volumes described above). The criteria for achieving a sensitive WIMP dark matter search was set as requiring:

- a 50-100 keV nuclear recoil threshold;
- $\mathcal{O}(10)$ background events;
- $\mathcal{O}(100)$ photons detected per event to allow pulse shape discrimination (PSD).

Assuming that 1250 photons per 100 keV of prompt scintillation light is emitted (as measured by SCENE for 500 V/cm fields [75]), the studies in Section 3.3 show that reaching 100 photons per event is realistic. A pseudo-Monte Carlo simulation of the Poisson distributed light output was performed to determine the width of a typical PSD variable f_{90}, defined as the ratio of light detector detected in the first 90 ns of an event to the light detected in the second 90 ns of an event, for the ^{39}Ar decays. The results taken from measurements from the SCENE experiment [75]. Reference [10] shows PSD is expected to reach the 10^{10} rejection level for electron/gamma backgrounds. We also direct the interested reader to the PSD study [76] to suppress the ^{39}Ar decay background in DEAP.

All other electron/gamma backgrounds are expected to be subdominant to the ^{39}Ar and will thus be removed by PSD. Neutron backgrounds were managed as described above. The main background will be from irreducible atmospheric neutrinos at the so-called neutrino floor. A full background table from the study is shown in Ref. [10].

The background rates are used to set a 90% C.L. sensitivity to WIMP dark matter Ref. [10] shows that a three-year search with this detector will have comparable sensitivity to planned next-generation detectors, which have expected run times of 10 years. This timescale allows a rapid cross-check of any signals discovered in these detectors, in particular for the liquid argon experiments such as DarkSide-20k [9] or ARGO [77].

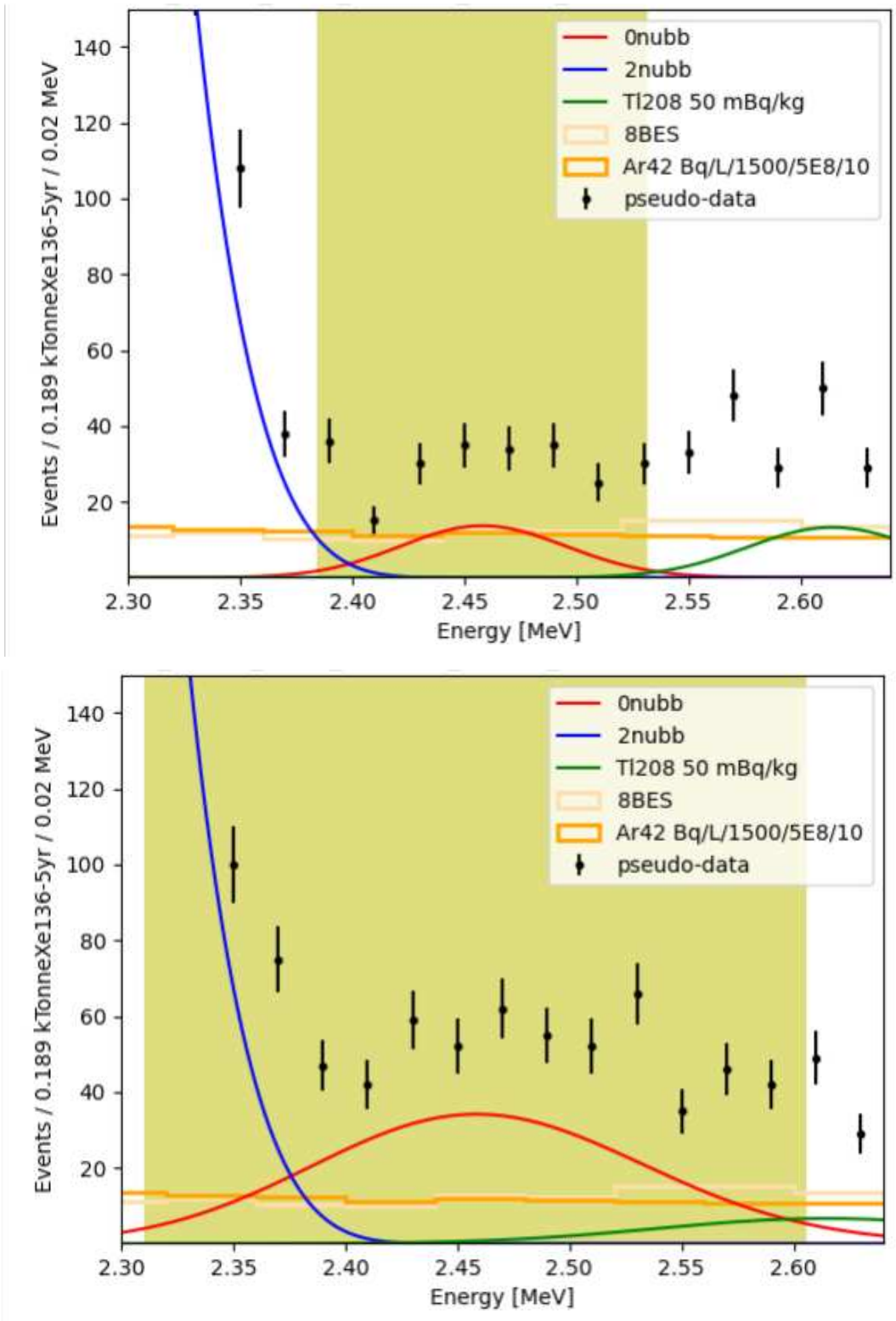


Figure 13. An optimistic background/resolution scenario for a ^{136}Xe $0\nu\beta\beta$ half-life of $5{\cdot}^{28}$ years. Detector energy resolution is 1.5% on top. Backgrounds are as discussed in the text. On the bottom is the same with a reduced resolution of 3% and for a half-life of 1E28 years.

3.7.1. Seasonal Variation of Rate for WIMP Dark Matter The prominent model for dark matter (DM) is the so-called Standard Halo Model (SHM) [78] featuring WIMP DM. The SHM describes a basic isometric spherical distribution of WIMP DM around our galaxy and has been used due to it being a good trade-off between realism and simplicity. The relative velocity of our solar system of 233 km/s [79] at which the Sun moves through the gas-like halo of WIMP DM induces as a "WIMP wind". Figure 14 illustrates the Sun's rotation in our galaxy together with Earth's solar orbit into and out of the WIMP wind. We modeled the Sun's rotational and peculiar velocities into one combined effective velocity of 233 km/s in the galactic plane and accounted for the $60°$ inclined plane of Earth's orbit. We were then able to accurately describe the annual modulation of detectable WIMP rate R on Earth by one simple periodic sinusoidal function with one amplitude parameter A and a maximal rate on June 1 of each year[79] [80]:

$$R(\,[d^{-1}]\,) = A\,[d^{-1}]\;\times\;\cos\left(\frac{2\pi}{T[d]}\times(\,t[d]-t_{June1}[d]\,)\right)+R_{avg}\,[d^{-1}] \tag{2}$$

The constant term R_{avg} is the average annual rate. Earth's period T is 365.2422 days and the phase corresponding to the maximal rate R_{max} observable on June 1 of each year is $2\pi\cdot t_{June1}/T = 2\pi\cdot 0.415$.

Galactic WIMPs in the halo are assumed to have a Maxwell-Boltzmann-like velocity distribution. The effective WIMP velocity distribution is shifted up when Earth is flowing maximally into the WIMP wind and shifted down when Earth is flowing maximally with the WIMP wind. Due to this aspect, an analysis on the annual modulation of the detection rate R could provide a powerful tool for identifying WIMP DM. Due to the unrivaled large fiducial mass of 3 kTons and a potentially very long DUNE operation of one decade (or even several), this concept can offer a unique detection of the seasonal variation of the detectable WIMP rate R at a sufficient statistical significance for providing a smoking gun signature for the WIMP nature of DM. This would be particularly of interest in case upcoming generation-2 DM experiments like LZ [81] and/or XENONnT [82] have evidence for WIMPs near their sensitivity. It would be nearly impossible for the planned generation-3 DM experiments [83] [84] to make such a smoking gun detection proving the WIMP nature of DM.

The differential rate of interactions in an arbitrary detector for the SHM is described in Equation 3:

$$\frac{dR}{dE_R} = \sigma_N^{SI}\frac{A^2 m_A N_T \rho_\chi}{2m_\chi \mu_N^2}F^2(E_R)\int\limits_{\nu_{min}(E_R)}^{\infty}\frac{d^3\overrightarrow{\nu}}{\nu}f_\oplus(\overrightarrow{\nu},\overrightarrow{\nu_{obs}}) \tag{3}$$

where σ_N^{SI} is the spin-independent-nucleon cross-section for WIMPs, A is the atomic number of argon, m_A is the mass of argon, N_T is the number of target nuclei, ρ_χ is the local dark matter density ($0.3\frac{GeV}{cm^3}$), m_χ is the mass of a WIMP, μ_N is the reduced WIMP-nucleus mass, $F(E_R)$ is the the nuclear form-factor, ν_{min} is minimum WIMP velocity to produce a recoil of energy E_R, ν is WIMP velocity, and ν_{obs} is the observer velocity with respect to the galaxy. When Earth is

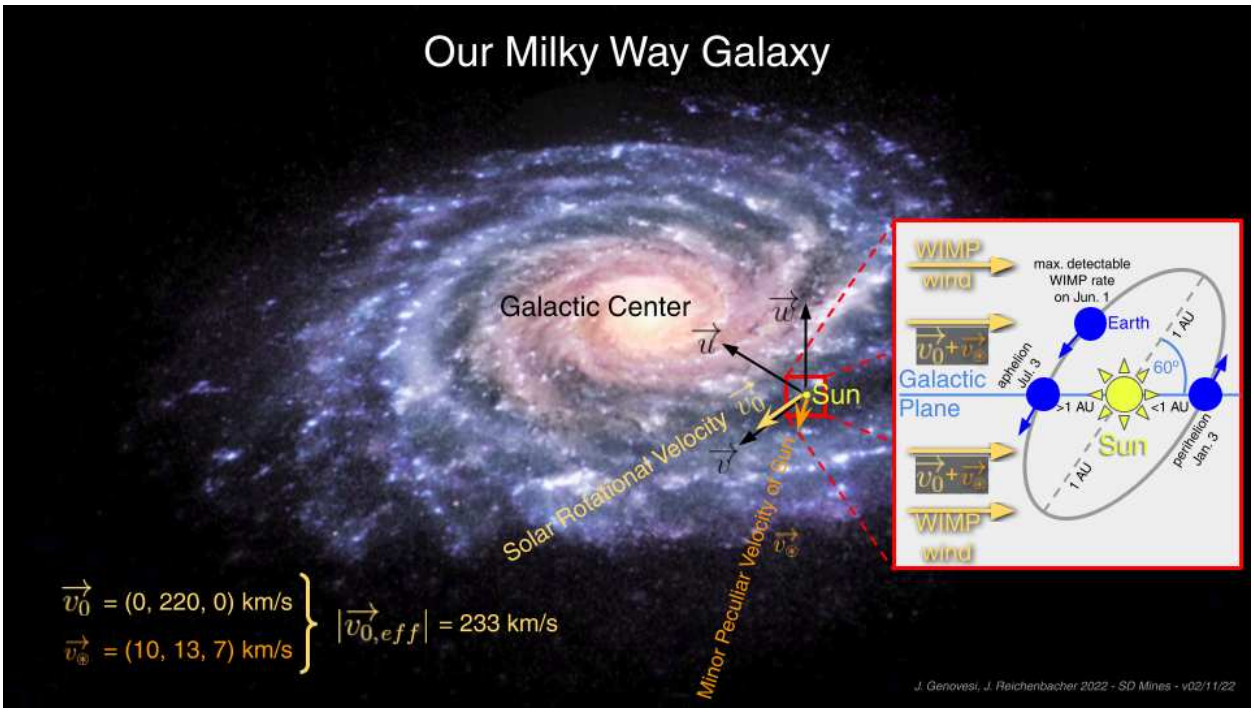


Figure 14. Galactic WIMP wind as it relates to Earth's orbital plane employing an illustrative rendition [79] of our Milky Way galaxy. Our solar system's velocity in reference to our galaxy has contributions from both a rotational aspect with a tangential velocity of 220 km/s and from a minor solar peculiar aspect with velocity components $(U, V, W) = (10, 13, 7)$ km/s [80]. In our model the combined relative velocity of our solar system is then 233 km/s at which the Sun moves through a gas-like halo of WIMP dark matter assumed to have a Maxwell-Boltzmann-like velocity distribution. This induces what we experience as a "WIMP wind". This WIMP wind is at an angle compared to Earth rotation around the Sun as pictured in the zoomed in diagram, with the effective WIMP velocity distribution shifted up when flowing maximally into the wind and shifted down when flowing maximally with the wind. Due to this aspect, an analysis of the annual modulation of detectable WIMP rate on Earth could provide a powerful tool for identifying the WIMP nature of dark matter.

moving into or with the WIMP wind, it affects the differential WIMP rate, which in turn would affect our $\overrightarrow{\nu_{obs}}$. For ease, we can define:

$$\int_{\nu_{min}(E_R)}^{\infty} \frac{d^3\overrightarrow{\nu}}{\nu} f_{\oplus}(\overrightarrow{\nu}, \overrightarrow{\nu_{obs}}) = \zeta(E_R), \tag{4}$$

where

$$f(\overrightarrow{\nu}) = \frac{1}{N}\left(e^{\frac{-\nu^2}{\nu_0^2}} - e^{\frac{-\nu_{esc}^2}{\nu_0^2}}\right), \tag{5}$$

and

$$f_{\oplus}(\overrightarrow{\nu}, \overrightarrow{\nu_{obs}}) = f(\overrightarrow{\nu} + \overrightarrow{\nu_{obs}}) \tag{6}$$

ht

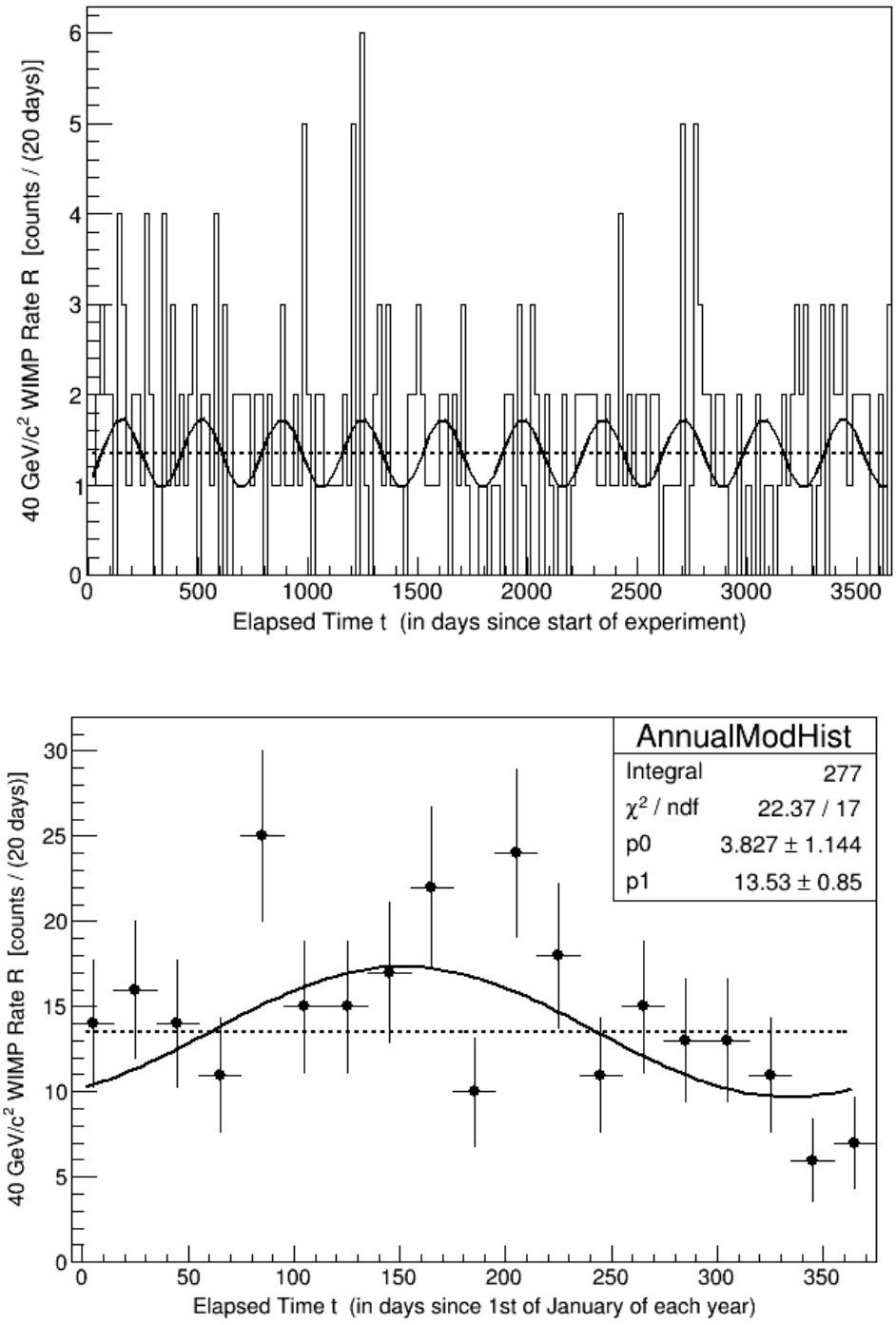


Figure 15. Example NEST[85] simulation of the annual modulation for $40\,\mathrm{GeV}/c^2$ WIMP dark matter with a cross section of $4 \times 10^{-48}\,\mathrm{cm}^2$ for a 10 year measurement time with 3 kTon LAr at $50\,\mathrm{keV}$ threshold. On top a Likelihood-fit result for a 10 year period and on bottom the same data combined into a single annual period starting on January 1 of each year fitted with a χ^2-method. The ideal case without background is assumed in this study.

with N as a normalization factor, ν_0 is the expected WIMP velocity, and ν_{esc} is the escape velocity of the galaxy. Using Equation 5, we can solve for individual cases of WIMPs in certain speed brackets.

As mentioned earlier, Earth's orbit can play a crucial role for differential WIMP rate predictions in the SHM. As the solar system travels through our galaxy, observers on Earth would observe WIMPs dominantly from a certain direction in a windshield-to-rain like effect. This is due to the distribution of WIMPs being treated as a gas with a Maxwell-Boltzmann-like velocity distribution with the stars in our galaxy moving through the dark matter due to their orbits around the galactic nucleus. In addition to the standard rotational contribution from the solar system, a minor peculiar velocity exists of our solar system traveling through the galaxy. This WIMP wind would be at an angle of 60° towards Earth's orbital plane. This means that during June 1, the Earth will experience a maximum effective flux of WIMPs while on December 1 the Earth will have experienced the lowest effective WIMP rate, as one can see again in Figure 14.

An example fit using Equation 2 of an annual modulation signal simulated in NEST [85] without background is shown in Figure 15 on top for a 10 year period, and on bottom the same data combined into a single annual period starting on January 1 of each year. This is for a $40\,\mathrm{GeV}/c^2$ WIMP with a cross section of $4 \times 10^{-48}\,\mathrm{cm}^2$, which is very close to the limit of sensitivity of the upcoming generation-2 xenon dark matter experiments LZ [81] and XENONnT [82]. We further assumed a $3\,\mathrm{kTon} \times 10\,\mathrm{year}$ exposure of our proposed low background LAr module with a $50\,\mathrm{keV}$ threshold resulting in 277 events. The bottom plot of Figure 15 shows a good χ^2-fit result for the amplitude $A = 3.827 \pm 1.144$ from Equation 2. It confirms the possible measurement of the seasonal variation of the WIMP rate at a sufficient statistical significance for providing a smoking gun signature for the WIMP nature of DM. This will be uniquely possible with this detector, due to the unrivaled large mass of $3\,\mathrm{kTons}$ vs. only 300 tons of argon for ARGO [84] and 100 tons for a Gen-3 xenon experiment [83]. Moreover, the annual modulation effect in xenon is significantly smaller due to the relatively lower energies of nuclear recoils in xenon compared to argon. Last but not least, the logistics of a decade long operation with this detector can utilize strong synergies with the DUNE long-baseline physics, including the cavern availability and occupancy.

3.8. Additional Topics

This module, with its unprecedented combination of low background and size, also can explore several other topics. We describe several examples in this section.

3.8.1. Atmospheric neutrinos The detector will measure approximately 10 CEνNS events due to atmospheric neutrinos. These events have not yet been observed and this would allow a cross-check of background rates from the upcoming generation of dark matter experiments.

3.8.2. Strangelets Recently, the paper "Can strangelets be detected in a large LAr neutrino detector?" [86], predicted that a LArTPC detector is able to detect and discriminate light strangelets with (Z, A) between (2,14) and (7,70) for energies up to 10 GeV in the presence of radioactive background found at the surface. When operated underground the detection limits are expected to be extended due to lower background levels and, combined with the increased dimensions of the detector module, will improve the event rates with a factor of 40. In the case of strangelets the main uncertainties are due to the estimations of their survival probability deep underground. The presence of ^{39}Ar masks both ionization and scintillation signals from strangelets and induces false signals in the collected charge from ionization. The use of underground argon in this module allows a cleaner detection signal.

3.8.3. Charged micro-black holes and Superheavy dark matter Hawking [87] suggested that unidentified tracks in the photographs taken in old bubble chamber detectors could be explained as signals of gravitationally "collapsed objects" (μBH). The small black holes are expected to be unstable due to Hawking radiation, but the evaporation is not well-understood at masses of the order of the Planck scale. Certain inflationary models naturally assume the formation of a large number of small black holes [88] and the generalized uncertainty principle may indeed prevent total evaporation of small black holes by dynamics and not by symmetry, just like the hydrogen atom is prevented from collapse by the standard uncertainty principle [89]. Given the profound nature of the issues addressed, some disagreement and controversy exist.

In principle the direct detection of charged micro black holes with masses around and upward of the Planck scale (10^{-5} g), ensuring a classical gravitational treatment of these objects, is possible in huge LAr detectors. It has been shown that the signals (ionization and scintillation) produced in LAr enable the discrimination between micro black holes (with masses between 10^{-5} - 10^{-4} grams, and velocities in the range 250 - 1000 km/s) and other particles [90]. It is expected that the trajectories of these micro black holes will appear as crossing the whole active medium, in any direction, producing uniform ionization and scintillation on the whole path.

Along these lines, an analysis looking for multiple co-linear nuclear recoils can also probe ultra heavy dark matter beyond the Plank scale, as described in Ref.[91, 92]. Sensitivity to the heaviest dark matter candidates is limited by the number density of the dark matter, which is inversely proportional to the mass, as the ability to detect heavy dark matter with a high cross section is set by the probability that a dark matter particle enters the detector. As such, sensitivity to the highest masses scales with the detector's surface area, and would leverage the large size of this module compared to DEAP-3600, which has a 1.7 m diameter.

Similarly, in the direct detection of the charged micro-black holes, unlike in traditional WIMP detection, there will exist both ionization and scintillation signals from direct interactions and from recoiling nuclei. The capability to perform pulse shape discrimination in this detector will allow these tracks to be identified. Natural radioactivity is the main source of background in this case and the reduced number of free electrons (and photons) from beta decays of ^{39}Ar will allow a

significant improvement of the capability of the detector to correctly identify the micro-black hole signals.

3.8.4. Other topics This detector would have sensitivity to a small number of CEνNS events within the neutrino beam. It may have applications to geologic tomography. The improved energy resolution for low energy could have applications in searches for other exotics and beyond the standard model physics phenomena. Though the optimal search region for a diffuse supernova neutrino background is above the energy of the solar neutrinos [93], and thus the radioactive backgrounds, the improved energy resolution of this detector will again likely improve the search sensitivity.

4. Conclusion

We have presented a design in this paper for a low background kTon-scale LArTPC to potentially expand the current physics program for such detectors. The design is based on the vertical drift detector planned for DUNE's second far detector module. The module discussed is a candidate for a third or fourth DUNE "Module of Opportunity." It is realized by providing additional shielding, stringent radioactive background control and enhanced light detection to the nominal vertical drift module. Energy resolution will benefit in all energy ranges due to event reconstruction and topology classification improvements from the superior light detection system and the quiet detector, which will allow to capture more cascade gammas[13] and thereby improve the hadronic component of neutrino-nucleus interactions.

The physics goals achieved by the SLoMo design extend the capability of large LArTPCs to search for solar and supernova neutrinos, neutrino-less double beta-decay, and WIMP dark matter. At the same time the design proposed here, by the nature of its small perturbations to the vertical drift module, assures continuing strong support to the long-baseline neutrino oscillation program to measure remaining parameters in the PMNS matrix.

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